Exploring Natural Language Prompts for Privacy Settings in VR

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Abstract

Interfaces that use large language models (LLMs) enable users to express their preferences in natural language, eliminating the need to navigate complex menus. In virtual reality (VR) applications this approach could help to simplify setting privacy settings. To explore this potential, we conducted an exploratory workshop in a co-design setting using a Wizard-of-Oz prototype. We analysed participants' sentiments and identified their primary concerns regarding the trustworthiness, control and usability of voice-based privacy systems. Based on these findings, we propose three areas that we plan to investigate further: (1) providing VR-native verification and adjustment options, (2) offering contextual information to users when formulating a prompt, and (3) aligning system-level architecture with users' needs.

CCS Concepts

• **Security and privacy** → **Usability in security and privacy**; *Privacy protections*; • **Human-centered computing** → **Virtual reality**; **co-design design**.

Keywords

Usable Security and Privacy, VR, Co-Design, Large Language Models

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1 Introduction

For most of digital computing's history, the dominant interaction paradigm has been command-based. Users issue specific commands to the computer through either command-line interfaces or graphical user interfaces (GUIs). These interfaces facilitate a back and forth between the user issuing a command and the system responding to it. They work well where only small amounts of communication between the user and the system are required, but effectively limit users' interaction options to the finite set of commands that the user interface (UI) gives them access to. Of course, integrating more flexible interaction options adds complexity to the UI.

In the field of virtual reality (VR), the limitations of command-based interfaces, combined with the limited input modalities – mostly button inputs and hand or controller tracking – restrict users' agency when interacting with complex interfaces, such as privacy and security settings. In contrast, systems built around natural language processing (NLP) tools like large language models (LLMs), enable users to communicate with digital systems using natural language, providing greater flexibility in how they instruct the system. Such systems could foster a new interaction paradigm, which Nielsen [15] calls intent-based outcome specification, and which we will refer to as intent-based for brevity.

In order to gain a better understanding of the opportunities and challenges that intent-based systems entail in the context of privacy settings, and as a starting point for further research in this area, we conducted an exploratory pilot study using a co-design methodology. To this end, we created a Wizard-of-Oz prototype that adapts traditional privacy settings interfaces by introducing an LLM interaction. We then ran a co-design workshop to engage users in the early-stage design process and gather their feedback and opinions. Participants were recruited from the queer community, which is a marginalised group with an increased need for effective privacy controls [20].

While participants were generally sceptical of voice interactions due to privacy concerns over their voice data, they also recognised the concept's potential and expressed a willingness to use it if their concerns were addressed. Further analysis of the data gathered during the workshop, we identified three key challenges to the future development of intent-based systems, for which we suggest three opportunities for intervention that we plan to investigate further in upcoming follow-up studies.

2 Related Works

With increasingly capable machine learning models, like LLMs, new interaction paradigms could be enabled, significantly changing how humans interact with digital systems. Vemuri [24] describe the potential of LLM-based systems to increase personalisation, efficiency, and accessibility. They also highlight that current implementations still only use a limited set of interaction paradigms, most prominently chat interaction, which dates back to the oldest systems that enable user interaction through natural language, like the one by Weizenbaum [25].

Fast et al. [9] implemented a conversational model for data science workloads which uses a statistical model to identify the task requested by the user. In their study users had concerns about the

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system's transparency, prompting the authors to implement additional control opportunities. Such models are sometimes referred to as intent-based, for example by Zylowski et al. [27], as they map users' input to a finite set of predefined intents. To differentiate these approaches from the intent-based paradigm discussed here, we will refer to them as non-LLM-based.

A comparative approach was used by Zylowski et al. [27] to contrast a form-based GUI with a non-LLM-based and an LLM-based conversational agent (CA). While the non-LLM based CA was perceived as functionally on par with the GUI, the lack of advanced conversational capabilities resulted in a lesser perceived degree of personalisation and consequently competence and autonomy. However, this did not translate to an increased level of trust in the system, which the authors potentially attribute to the disconnect between detailed conversations and the generic results provided by the system. Similarly, research on voice user interfaces (VUIs) has mostly focused on conversational systems, typically in the form of personal assistant systems. A significant focus in this area is placed on users' trust in the systems [6, 26], with Pal et al. [18] distinguishing between emotional and cognitive trust – the perceived intelligence of the system having a significant impact on users' cognitive trust. Importantly, existing research emphasises the additional privacy concerns associated with using voice as an interaction modality, with users fearing the misuse of collected voice data [5, 26].

These systems can still be understood as command-based interfaces, as described by Nielsen [14]. Developing LLM-based systems that go beyond this paradigm requires a better understanding of users' interactions with LLMs. Subramonyam et al. [22] refer to the process by which a user communicates their intent to an LLM as envisioning and identify prompt formulation and a lack of understanding of the LLM's system model as two primary challenges that create a high cognitive demand for users.

The immersive nature of virtual reality (VR) requires sensors to capture more sensitive data than other technologies, creating unique privacy risks [10, 21]. As VR applications rely on continuous tracking, avoiding this data capture would diminish core functionality [10]. Spatial and inertial data can reveal physical traits such as height, arm length and health, which can be used to infer sensitive data like users' gender [11, 12]. Additionally, behavioural patterns – including head, hand, and gait movements – enable re-identification across users, even with limited available data such as height and arm length [11–13, 19]. Beyond movement, eye-tracking and health sensors can expose gaze, facial features, emotions, and medical conditions, although these modalities are not considered in this project [12].

These issues are compounded for marginalised communities, who face unique privacy challenges when using digital services. Queer people, often referred to as LGBTQI+, are particularly affected, facilitating a need for effective and usable privacy tools [20], not just on social media platforms but also when interacting with digital systems in general Ovalle et al. [17]. This applies especially to sensor data that can be used to infer characteristics such as gender and walking style. As previously discussed, these types of data are necessarily collected in large quantities by VR applications, exacerbating the resulting issues for this user group. Although there has been an increase in HCI research activity regarding the needs of queer communities, Taylor et al. [23] highlight that these groups are still often treated as an afterthought.

3 Methods

To explore the concept of intent-based interfaces for privacy settings in VR, we conducted a co-design workshop ($N = 5$). The workshop aimed to gather user attitudes towards intent-based systems, insights into user expectations regarding functionality, and suggestions for future designs.

In line with a constructive, prototype-first approach within HCI research [16], a VR prototype was implemented on a Meta Quest 3 head-mounted display (HMD) to gather user prompts and real-time feedback when users were interacting with an intent-based system. Users first complete a training experience to familiarise them with the general concept of VR applications, in which they are placed in a small, village-like environment and asked to collect various objects. This was implemented to account for differences in participants' previous exposure to VR applications and to ensure users had a basic knowledge of the tracking data that enabled VR interactions, such as hand, position and head tracking.

First, information on the study was provided, followed by a word association exercise centred on privacy, security and VR; to make sure that all key terms were understood. Afterwards, users tried out the Wizard-of-Oz prototype [4]. This setup enabled us to demonstrate the functionality of an intent-based system and engage participants in early co-design without extensive implementation effort. Users first engage with a menu informing them about the application's gathering of telemetry data and asking them to speak a prompt to communicate their privacy preferences. A researcher simulated the LLM system by transcribing user prompts in real time and mapping them to predefined options on paper as they were spoken. The available options reflected the current permission model of Meta Quest HMDs, as well as additional options based on Abraham et al. [1]. After speaking their prompt, participants left the VR application and reviewed the transcribed prompt and resulting mappings. They were given the option to make adjustments if they felt their prompt should have resulted in different settings. A subsequent group discussion allowed participants to share their experiences and opinions on the prototype tested. In the final segment, participants took part in a co-design design session in two subgroups to discuss ways to enhance the prototype.

Participants were recruited from the queer community through researchers' personal networks. Participants were offered complimentary catering in exchange for their participation. All applicable ethics clearance processes were followed, and approval was obtained. The workshop structure and materials were designed in close collaboration with a researcher who is a member of the queer community to ensure it would not expose participants to additional social or privacy risks.

The participants were predominantly in the 30–40 age cohort ($N = 4$), with one participant in the 20–30 age cohort. Of the participants, two self-identified as female, two as non-binary, and one as male. They self-reported as having relatively little prior experience with VR applications ($\bar{x} = 1.6, \sigma = 1.34$; all on 7-point Likert scales), and somewhat higher experience with prompting AI-tools ($\bar{x} = 2.6, \sigma = 2.07$) and a heightened sensitivity to data

privacy in general ($\bar{x} = 5.8, \sigma = 0.84$). All participants had normal or corrected vision. The workshop was conducted in German and audio recordings were made of the entire workshop that were later transcribed using noScribe Software [7] and manually revised. Based on these transcripts, an inductive thematic analysis was performed independently by two experts and then consolidated through discussion [3].

4 Results

Three primary codes of interest emerged during the analysis: control, usability, and trust.

Trust. While participants expressed some amount of trust in the system choosing appropriate settings from their prompt, a fundamental issue was their general distrust of providing voice data to application vendors, especially in the context of gender detection and transition, with one remarking "[...] Tonaufnahme fühlt sich nicht wie Kontrolle an" [Voice recordings don't feel like being in control]. They agreed that having voice data processed on-device only would help alleviate that to an extent. Some concerns were additionally raised over the accuracy of speech to text functionality in general, with participants worried that they would be misunderstood.

These concerns were partially alleviated by the system offering the option to review and adjust the prompt's results, with one participant describing it as "durch spätere Möglichkeit zur Kontrolle [...] sehr transparent" [very transparent, due to the later option to double check]. At the same time, participants felt that providing this option made the prompt somewhat "redundant".

Usability. Some participants found the modality of voice input in VR to be awkward, with one describing it as "Einschüchternd aber komfortabel" [intimidating but comfortable], while others found it to be preferable, especially for people new to VR or when compared with more complex VR GUIs, SteamVR being named as an example.

Formulating a prompt in one go and with only limited contextual information was perceived as demanding and unusual. One participant described that they "nicht wirklich darunter vorstellen konnte, was jetzt genau der von mir will" [didn't really know what the system is expecting from me], while another said that initially, speaking the prompt freely made them feel insecure.

Control. This was also reflected in participants making relatively short prompts ($\bar{x} = 20.4$ words), with one remarking they wondered "was sage ich denn jetzt?" [what do I say now?], leading to participants not feeling like they made an informed prompt. They were concerned that their prompt was not comprehensive enough, which could result in the system making choices on their behalf that they did not explicitly want. The lack of guidance during the prompting process was however also perceived as an opportunity, with participants reporting that it forced them to actively think about their preferences, thus providing "mehr Möglichkeiten der Selbstbestimmung" [more options for autonomous decisions].

During the prompt exercise and in the subsequent discussions, participants focused intensely on who would collect their data and for what purpose, with one participant's prompt allowing for the gathering of data as long as it was not shared with third parties. This was in contrast to the permission model that the prototype was based on, which focused on what types of data would be collected, resulting in participants seeing a divergence between their prompts and the resulting permission model. Participants deemed it crucial across the board that a way to check and adjust the results of their prompts was provided to ensure correct outcomes.

5 Discussion

Participants were open to the concept of intent-based interactions and saw distinct advantages. However, they also raised a number of concerns based on which we discuss three overarching interventions.

First, the results highlighted that the implementation used in our prototype created a conflict between users' mental models and system architecture, with the former focusing mostly on who and for what purposes their data would be shared, while the system focused on what types of data would be collected. This resulted in participants feeling as though their prompt had not been processed accurately. While this can be partially attributed to the specific implementation used in the workshop, it highlights a more general conflict: If the greater flexibility that intent based systems can offer is not accommodated in the system architecture, answers will be too generic and appear disconnected from the user input, resulting in user expectations not being met. This in turn can significantly diminish users' trust in the system [18, 27].

A flexible system that can dynamically generate settings based on users' prompts, rather than being limited to a small set of options, could help alleviate this issue. Therefore, we propose that intent-based systems have an underlying system architecture that aligns with users' mental models, for example by providing rule-based systems over a static list of options or by implementing a system capable of learning from users' preferences to suggest options that meet their needs. This corresponds to intervention opportunity (1) in Figure 1.

Secondly, participants highlighted that having the opportunity to verify and modify the resulting settings after the prompt, was essential to ensure a level of confidence and control. However, they also questioned the utility of the interaction if they face an extensive command-based interface afterwards. While the demand for these control opportunities will likely decrease as users get more experienced with these systems [22] and may be lower from the onset for other user groups, it is very likely that some level of control opportunities will be required, as this is also observed in research on conversational agents [9].

To avoid making users feel like they are going through the same settings twice, intent-based systems will therefore need to strike a balance between providing users with enough intervention and control options, while also keeping the entire process usable and optimised for low interruption lag, reflected in intervention opportunity (2) in Figure 1. Visualisations have been explored in the realm of privacy [2] and could provide users in an immersive VR setting, with an intuitive way to spatially interact with the results of intent-based systems. We plan to explore this in future work.

Finally, participants consistently reported that the relatively small amount of contextual information provided before speaking the prompt, led to them feeling intimidated or overwhelmed when speaking the prompt. As this seems indicative of a high cognitive

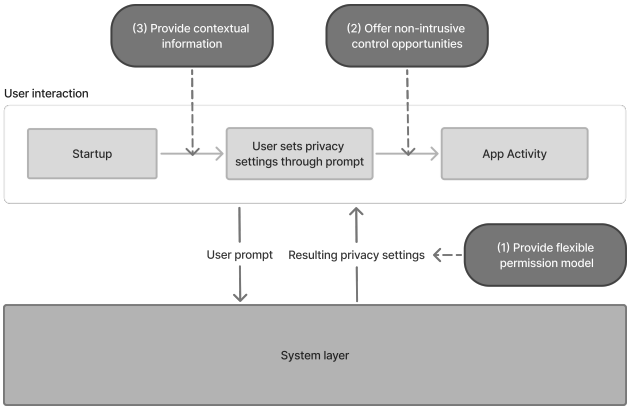

Figure 1: A simplified user flow that highlights opportunities to increase (1) trust, (2) control and (3) usability.

load, we plan to explore this further in future work. Furthermore, given the relatively high level of technological literacy the group displayed otherwise, technical difficulties seem to be an unlikely cause and this is more likely attributed to a general increase in cognitive demand associated with formulating prompts, as reported in prior work [22]. From a usable security and privacy (USEC) perspective, this resembles findings on command-based interfaces, where users seeking secure and private systems must recall precise syntax, commands, and parameters without contextual guidance, resulting in high cognitive load and error-prone interaction. This aligns with prior work highlighting the limited usability of command-based interfaces [24].

Participants did not perceive this phenomenon as purely negative, but also reported a heightened feeling of agency. Leveraging this, while maintaining a usable system for users, could be done through the scaffolding of contextual information that has been proven valuable in complex interactions to encourage user engagement (with privacy and security) while offering sufficient usability. Prior research has also emphasised this potential, for instance, through systems such as dynamic prompt middleware [8], which is consistent with our findings.

Subramonyam et al. [22] have conceptualised the cognitive challenges associated with LLM-based interaction, which are in line with our results, and proposed design guidelines to address these challenges. However, their emphasis on identifying an optimal balance between prompting and restraint introduces a tension when considered in the context of USEC, where privacy and usability must be carefully negotiated. Thus, future intent-based interfaces must balance the provision of sufficient contextual information in a non-intrusive manner with preserving user agency—an open challenge reflected in intervention opportunity (3) in Figure 1.

Limitations. Recruiting participants from this marginalised community is challenging due to widespread distrust of technology, particularly around privacy and security [20]. Accordingly, participants were recruited through one experimenter's personal network, providing a trusted context for this sensitive study. While this approach was intentional, aimed at engaging the community early and avoiding their frequent marginalisation in later-stage research Taylor et al. [23], it may limit the generalisability of our findings. Importantly, both researchers and participants agreed that the work should not be framed as a queer-focused study. Instead, the community is engaged as one with heightened privacy and security requirements, establishing a deliberately stringent baseline for future research in this domain.

A researcher mapped participants' prompts to a set of privacy options in real time and on paper. While this enabled us to mimic an LLM, it created a break in the immersive experience, which may have impacted their subsequent responses.

6 Conclusion

In a co-design workshop, we explored the concept of an intent-based interface that allows users to specify their privacy preferences through a natural language prompt using a Wizard-of-Oz prototype of a VR application. We identified three primary challenges that participants had with the concept: a mismatch between the user and system layers of our application; a strong demand for opportunities to control outcomes; and heightened cognitive load when formulating a prompt resulting in diminished usability. To address the mismatch between the user-facing interaction and the system layer, we propose three high-level interventions, which we plan to explore in future work. With this, we present a first exploration into intent-based interaction for privacy and security settings and propose a high-level approach to address some of the key challenges that emerged through this study.

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DeepL Write was used to make corrections and suggest alternative phrasing. ChatGPT was used to generate supplementary materials for the thematic analysis. All AI-generated content was carefully reviewed and revised before inclusion.

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A Provided Privacy Options

Option (German)	English Translation
Based on Meta Quest Permission Model	
Kamera	Camera
Fotos und Video	Photos and Videos
Mikrofon	Microphone
Musik und Audio	Music and Audio
Räumliche Daten	Spatial Data
Speicher	Storage
Standort	Location
Additional Options Provided	
Profiling: Erstellen eines Profils anhand von Annahmen über demografische Merkmale	Profiling: Creating a profile based on assumptions about demographic characteristics
Zu Werbezwecken: Du bekommst Werbung gezeigt, die zu den Interessen deiner demografischen Gruppe passt	For advertising purposes: You will see ads that match the interests of your demographic group
Zur Anpassung der Spielerfahrung: Deine Spieleinstellungen (z. B. Größe und Aussehen von Menüs) werden an die Vorlieben deiner demografischen Gruppe angepasst	To adapt the gaming experience: Your game settings, such as the size and appearance of menus, are adjusted according to the preferences of your demographic group
Profilbildung anhand des folgenden Merkmals: Alter	Profiling based on the following characteristics: Age
Profilbildung anhand des folgenden Merkmals: Geschlecht	Profiling based on the following characteristics: Gender
Profilbildung anhand des folgenden Merkmals: Fitness	Profiling based on the following characteristics: Fitness
Profilbildung anhand der folgenden Merkmals: körperliche Einschränkungen	Profiling based on the following characteristics: Physical impairments
Biometrische Identifikation	Biometric identification
Biometrische Identifikation: Verwendung des automatischen Logins	Biometric identification: Automatic login
Biometrische Identifikation zu Werbezwecken: Du bekommst Werbung gezeigt, die zu deinem bisherigen Verhalten passt	Biometric identification for advertising purposes: You are shown ads that match your previous behavior
Biometrische Identfikation zur Anpassung der Spielerfahrung: Deine Spieleinstellungen (z. B. Größe und Aussehen von Menüs) werden an dein bisheriges Verhalten angepasst	Biometric identification to adapt the gaming experience: Your game settings, such as the size and appearance of menus, are adjusted according to your previous behaviour.

Table 1: The options that were used in the workshop to map privacy prompts to.

B Prototype Screenshots

Figure 2: Screenshots from the prototype used. From left to right: Users engage in a collection game; the system introduces users to the prompt; users can speak the prompt to the system.

C Prompts spoken by Participants

Participant	Prompt	English Translation
P1	Keine Daten sammeln.	Collect no data.
P2	Ich möchte, dass nur so viele Daten aufgenommen werden, wie unbedingt nötig ist, um dieses Spiel zu spielen.	I want only as much data to be recorded as is necessary to play this game.
P3	Die Bewegungsdaten dürfen umfassend verwendet werden, ohne Einschränkungen.	Movement data may be collected without restrictions.
P4	Ich hätte gerne, dass meine Daten nicht von Dritten und am besten auch nicht von irgendwelchen Bearbeitern gesehen werden können. Und, dass sie zwischen mir und dem Headset sind, um Spiele optimal laufen lassen zu können. Und dann, wenn ich das möchte, sollten sie einfach zu löschen sein.	I want my data to not be shared with third partys, including editors. I also want them to stay between me and the device, only for the purpose of having the game run optimally. The data I provide should be easy to delete.
P5	Es ist okay für mich, dass Telemetriedaten erfasst werden, egal welche, solange sie nur zur Verbesserung meines Spielerlebnisses beitragen und nicht an Dritte weitergegeben werden.	It's okay that any kind of telemetry data is collected, as long as it is used to improve my gaming experience and is not shared with third parties.

Table 2: Prompts spoken by participants in the Wizard-of-Oz prototype. Provided are original German quotes and translations.